

Experiment 10:

Step 1: Create Evidence Folder

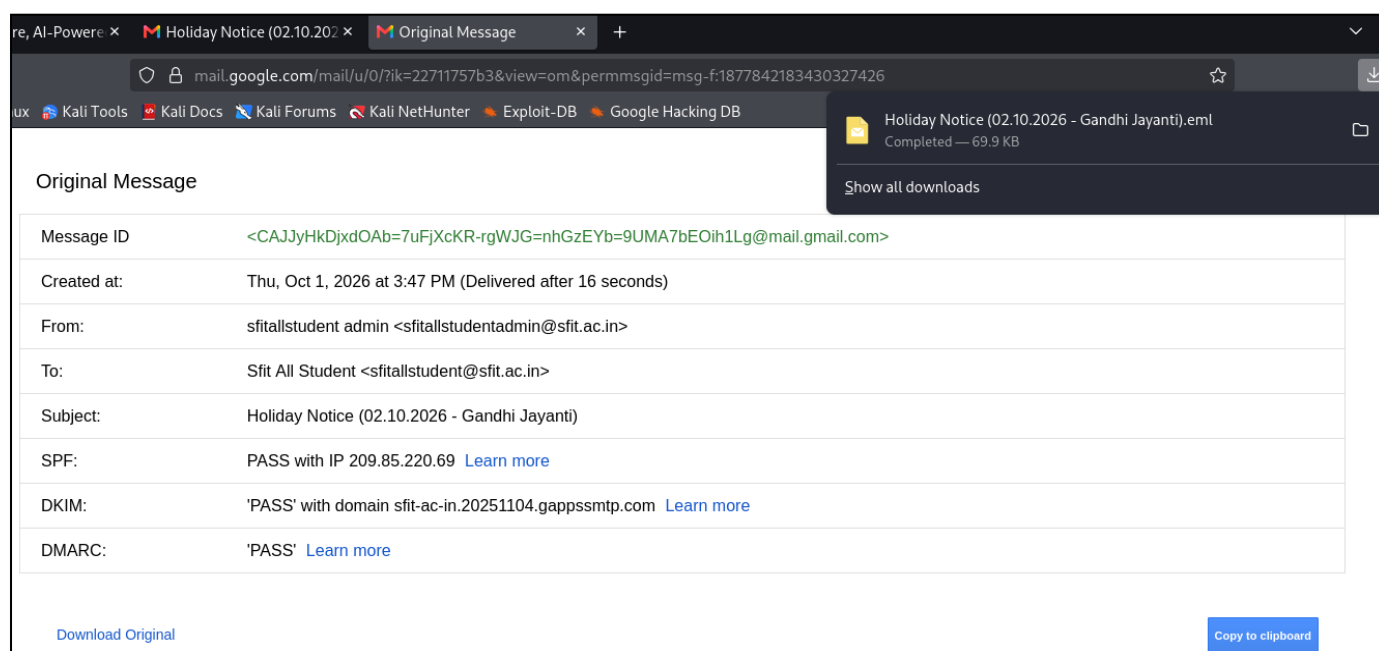
1.1 Create Case Directories and Enter the Working Directory

```
(student@Kali)-[~/Shivam]
$ mkdir -p email_forensics/{evidence,analysis,reports}

(student@Kali)-[~/Shivam]
$ ls
email_forensics  network_capture.pcapng
```

```
(student@Kali)-[~/Shivam]
$ cd email_forensics
```

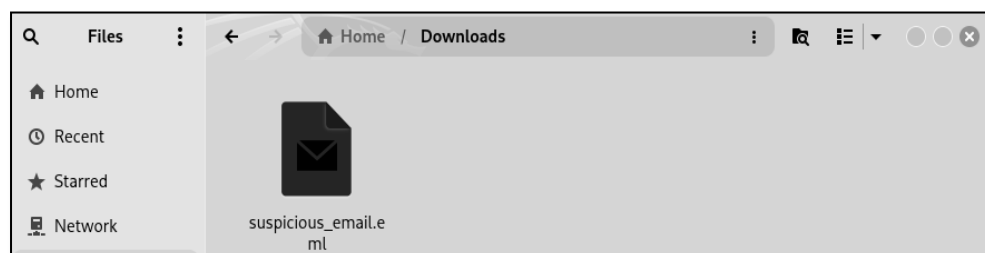
1.2 Locate the Original Email File



Original Message

Message ID	<CAJJyHkDjxdOAb=7uFjXcKR-rgWJG=nhGzEYb=9UMA7bEOih1Lg@mail.gmail.com>
Created at:	Thu, Oct 1, 2026 at 3:47 PM (Delivered after 16 seconds)
From:	sfitallstudent admin <sfitallstudentadmin@sfit.ac.in>
To:	Sfit All Student <sfitallstudent@sfit.ac.in>
Subject:	Holiday Notice (02.10.2026 - Gandhi Jayanti)
SPF:	PASS with IP 209.85.220.69 Learn more
DKIM:	'PASS' with domain sfit-ac-in.20251104.gappssmtp.com Learn more
DMARC:	'PASS' Learn more

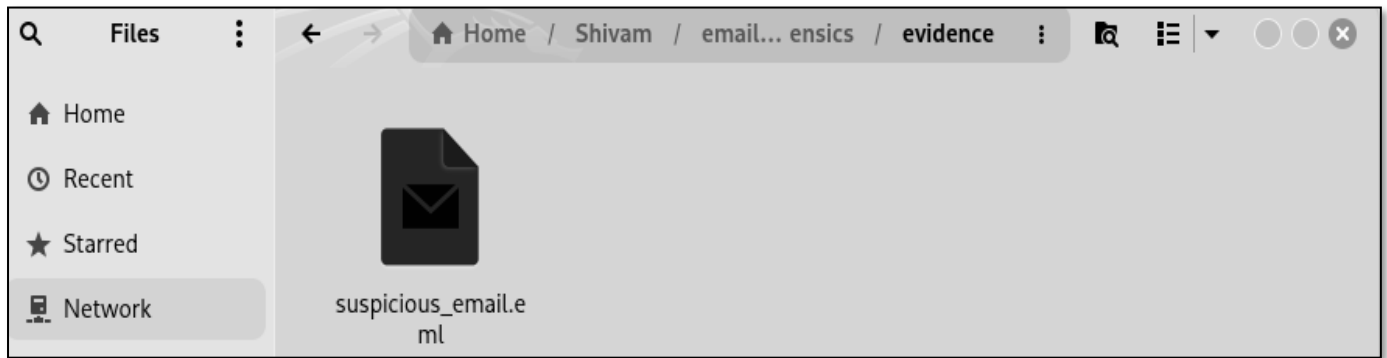
[Download Original](#) [Copy to clipboard](#)



1.3 Copy the Email File into the Evidence Folder

```
(student@Kali)-[~/Shivam/email_forensics]
$ cp ~/Downloads/suspicious_email.eml evidence/
```

1.4 Verify the Evidence File Placement



```
(student@Kali)-[~/Shivam/email_forensics]
$ ls evidence/
suspicious_email.eml
```

Observation:

- The mkdir -p command creates the email forensics directory structure containing the evidence, analysis, and reports directories.
- The cd email_forensics command changes the terminal's working directory to the email forensics folder.
- The original email is opened in the mail interface, where its message details and authentication statuses are visible.
- The suspicious_email.eml file is visible in the Downloads folder before being copied.
- The cp command copies suspicious_email.eml from Downloads into the evidence directory.
- The file manager shows suspicious_email.eml inside the evidence folder.
- The ls evidence/ command confirms that the email evidence file is present in the directory.

Step 2: Calculate Evidence Hash

2.1 Calculate the SHA-256 Hash

```
(student@Kali)-[~/Shivam/email_forensics]
$ sha256sum evidence/suspicious_email.eml
370005e27b26412450edcb87764d281164c84dd5e30427d120727d10ef1f6ba2  evidence/suspicious_email.eml
```

2.2 Save the Hash to the Report File

```
(student@Kali)-[~/Shivam/email_forensics]
$ sha256sum evidence/suspicious_email.eml > reports/original.txt
```

```

Open ▾  📄 original.txt
~ /Shivam/email_forensics/reports

1 370005e27b26412450edcb87764d281164c84dd5e30427d120727d10ef1f6ba2  evidence/
   suspicious_email.eml

```

Observation:

- The sha256sum command calculates the SHA-256 hash of the original email evidence file.
- The hash is saved in reports/original.txt, and the file is opened to verify the recorded hash.

Step 3: View Raw Email

```

~(student@Kali)-[~/Shivam/email_forensics]
$ less evidence/suspicious_email.eml

Delivered-To: shivampimple29@student.sfit.ac.in
Received: by 2002:a05:6504:e081:20b0:314:537d:9a10 with SMTP id m1-n2csp2146006ltd;
      Thu, 1 Oct 2026 03:17:40 -0700 (PDT)
X-Forwarded-Encrypted: i=5; AKwUvBzNSLdOyKLIy7fGkAtucYpsEY0mdwZh2Jp3S1xq6kSE5/qj0y+2V05KhA/4HYClvZCAto
X-Received: by 2002:a17:903:3503:b0:2e1:24ec:df50 with SMTP id d9443c01a7336-2e2e499e1c7mr31679025ad.2
      Thu, 01 Oct 2026 03:17:40 -0700 (PDT)
ARC-Seal: i=4; a=rsa-sha256; t=1790849860; cv=pass;
      d=google.com; s=arc-20260327;
      b=U6gG/CGM9UpqY+pFC+lud+EyvRkXkNWRn7DHkwYMQadiuhnjd4KBrhxLHs5Nd0s7ta
      lPLB0QVntfwG5ntq8gsG4JsqxHwa/8q3EryFLngGymF9h1cR0VWerefMVD0x0OixGgsq
      fIXLwbHu6hWuuu5XDzsEntMrj90Fz5Wuh/y9VivIyVtJdDox1RvTEKcj+FPtA5ywnU1
      7TAWZaTUetivPHEGshY2z5hisoaLdxhG/j8AZXvVjfPB7WrF9/IgygDnmpytJLMg9vFq
      QtG5JB1mv5zUWF+x1hEEb3MlggMrOR4GgxjUzWqHOSDkSuStchw9P5caCZ6nYBIJ+n9h
      fN1A==
ARC-Message-Signature: i=4; a=rsa-sha256; c=relaxed/relaxed; d=google.com; s=arc-20260327;
      h=list-unsubscribe:list-archive:list-help:list-post:list-id
      :mailing-list:precedence:to:subject:message-id:date:from
      :mime-version:dkim-signature;
      bh=Kn+9OdW7nfDBP+3u+Uvrt+QGj0btHL0NxAek4CqeGf0=;
      fh=ATZkNC4xUndPiAgoALsRzJL1cQPxpRXp21XtwQUM08w=;
      b=VFF2S2N4dIwXiUqVli86dxXI3uxlJYj7vEs6Wg/RTcz+DaKNbKhDiSvJEjtqH+lw
      K6TSRI9VzSg9HNM35MH/y1PWLlQw95bgeqfpp5VW70l5aknnMNG2YSxQJZmW7LQJ0iB6
      wCnukIppq37uSS6wsLDhR61S85Ctc3FChBl9h//+s2yLMDdxisoiy7LBS3UDHSL8pbAb
      0ujSHJ0o21h2txJLUX/rSeHUOTIBOD0Fuqfzu0zksDOMMxl6640p50tHudMDm8MUJGGW
      wF793fJfHjv8EyYSnmk0iArbhipBR7XycKz3RsdK4uHGkAPKZA0pb64g/EMQJiFO9+a
      VG8Q==;
      dara=google.com
ARC-Authentication-Results: i=4; mx.google.com;
      dkim=pass header.i=@sfit-ac-in.20251104.gappssmtp.com header.s=20251104 header.b=DeGvhUmZ;
      arc=pass (i=3 spf=pass spfdomain=sfit.ac.in dkim=pass dkdomain=sfit-ac-in.20251104.gappssmtp.co
      m)
      spf=pass (google.com: domain of sfitallstudent+bncbc4opa4k7ukrbqyg7dkqmgqez4lmumq@sfit.ac.in de
      MGQE4ZLMUMQ@sfit.ac.in;
      dmarc=pass (p=NONE sp=NONE dis=NONE) header.from=sfit.ac.in;

```

Observation:

- The less command opens suspicious_email.eml for viewing in the terminal.
- The raw email displays header information, including delivery details, routing information, and authentication-related fields.

Step 4: Extract Important Headers

4.1 Extract the From Header

```
(student@Kali)-[~/Shivam/email_forensics]
$ grep -i "^From:" evidence/suspicious_email.eml
From: sfitallstudent admin <sfitallstudentadmin@sfit.ac.in>
```

4.2 Extract the To Header

```
(student@Kali)-[~/Shivam/email_forensics]
$ grep -i "^To:" evidence/suspicious_email.eml
To: Sfit All Student <sfitallstudent@sfit.ac.in>
```

4.3 Extract the Subject Header

```
(student@Kali)-[~/Shivam/email_forensics]
$ grep -i "^Subject:" evidence/suspicious_email.eml
Subject: Holiday Notice (02.10.2026 - Gandhi Jayanti)
```

4.4 Extract the Date Header

```
(student@Kali)-[~/Shivam/email_forensics]
$ grep -i "^Date:" evidence/suspicious_email.eml
Date: Thu, 1 Oct 2026 15:47:24 +0530
```

4.5 Extract the Message-ID Header

```
(student@Kali)-[~/Shivam/email_forensics]
$ grep -i "^Message-ID:" evidence/suspicious_email.eml
Message-ID: <CAJJyHkDjxd0Ab=7uFjXcKR-rgWJG=nhGzEYb=9UMA7bEOih1Lg@mail.gmail.com>
```

Observation:

- The From header displays the sender's name and email address.
- The To header identifies the intended recipient of the email.
- The Subject header displays the subject as “Holiday Notice (02.10.2026 - Gandhi Jayanti).”
- The Date header displays the email's recorded date and time as Thu, 1 Oct 2026 15:47:24 +0530.
- The Message-ID header displays the unique identifier associated with the email message.

Step 5: Examine Reply-To Header

Observation:

The command to examine the Reply-To header has not been performed, so its screenshot is not available.

Step 6: Examine Return-Path Header

```
(student@Kali)-[~/Shivam/email_forensics]
$ grep -i "^Return-Path:" evidence/suspicious_email.eml
Return-Path: <sfital@student+bncBC40PA4K7UKRBQXG7DKQMGQEZ4LMUMQ@sfit.ac.in>
```

Observation:

The Return-Path header displays the envelope sender address associated with the email.

Step 7: Analyze Received Headers

```
(student@Kali)-[~/Shivam/email_forensics]
$ grep -i "^Received:" evidence/suspicious_email.eml
Received: by 2002:a05:6504:e081:20b0:314:537d:9a10 with SMTP id m1-n2csp2146006ltd;
Received: from mail-sor-f69.google.com (mail-sor-f69.google.com. [209.85.220.69])
Received: by 2002:a17:902:e80d:b0:2cc:89b8:4246 with SMTP id
Received: from mail-sor-f41.google.com (mail-sor-f41.google.com. [209.85.220.41])
```

Observation:

The Received headers display the mail servers involved in email delivery, including Google mail servers and their IP addresses.

Step 8: Extract IP Addresses

```
(student@Kali)-[~/Shivam/email_forensics]
$ grep -Eo '([0-9]{1,3}\.){3}[0-9]{1,3}' evidence/suspicious_email.eml
209.85.220.69
209.85.220.69
026.10.01.03
209.85.220.69
209.85.220.69
209.85.220.69
209.85.220.41
209.85.220.41
209.85.220.41
026.10.01.03
209.85.220.41
209.85.220.41
209.85.220.41
```

```
(student@Kali)-[~/Shivam/email_forensics]
$ grep -Eo '([0-9]{1,3}\.){3}[0-9]{1,3}' evidence/suspicious_email.eml | sort -u
026.10.01.03
209.85.220.41
209.85.220.69
```

Observation:

- The extraction command identifies dotted numeric sequences in the email, including repeated occurrences of 209.85.220.69 and 209.85.220.41.

- The sort -u command removes duplicate entries. The output also contains 026.10.01.03, which appears date-like and should not be treated as a confirmed IP address without verification.

Step 9: Investigate IP Address

```
(student@Kali)-[~/Shivam/email_forensics]
$ whois 209.85.220.69

#
# ARIN WHOIS data and services are subject to the Terms of Use
# available at: https://www.arin.net/resources/registry/whois/tou/
#
# If you see inaccuracies in the results, please report at
# https://www.arin.net/resources/registry/whois/inaccuracy_reporting/
#
# Copyright 1997-2026, American Registry for Internet Numbers, Ltd.
#

NetRange:      209.85.128.0 - 209.85.255.255
CIDR:          209.85.128.0/17
NetName:       GOOGLE
NetHandle:     NET-209-85-128-0-1
Parent:        NET209 (NET-209-0-0-0-0)
NetType:       Direct Allocation
OriginAS:
Organization:  Google LLC (GOGL)
RegDate:       2006-01-13
Updated:       2012-02-24
Ref:           https://rdap.arin.net/registry/ip/209.85.128.0

OrgName:       Google LLC
OrgId:          GOGL
Address:        1600 Amphitheatre Parkway
City:           Mountain View
StateProv:      CA
```

Observation:

The WHOIS output associates the investigated IP address, 209.85.220.69, with Google's allocated IP address range and identifies Google LLC as the organization.

Step 10: Perform Reverse DNS Lookup

```
(student@Kali)-[~/Shivam/email_forensics]
$ nslookup 209.85.220.69
69.220.85.209.in-addr.arpa      name = mail-sor-f69.google.com.

Authoritative answers can be found from:
```

Observation:

The reverse DNS lookup resolves 209.85.220.69 to the hostname mail-sor-f69.google.com.

Step 11: Check SPF Status

```
(student@Kali)-[~/Shivam/email_forensics]
$ grep -i "spf" evidence/suspicious_email.eml
arc=pass (i=3 spf=pass spf-domain=sfit.ac.in dkim=pass dkdomain=sfit-ac-in.20251104.gappssmtp.com
spf=pass (google.com: domain of sfitallstudent+bncbc4opa4k7ukrbqyg7dkqmgqez4lmumq@sfit.ac.in des
MGQE4LMUMQ@sfit.ac.in;
Received-SPF: pass (google.com: domain of sfitallstudent+bncbc4opa4k7ukrbqyg7dkqmgqez4lmumq@sfit.ac.in
arc=pass (i=3 spf=pass spf-domain=sfit.ac.in dkim=pass dkdomain=sfit-ac-in.20251104.gappssmtp.com
spf=pass (google.com: domain of sfitallstudent+bncbc4opa4k7ukrbqyg7dkqmgqez4lmumq@sfit.ac.in des
MGQE4LMUMQ@sfit.ac.in;
spf=pass (google.com: domain of sfitallstudentadmin@sfit.ac.in designates 209.85.220.41 as permi
spf=pass (google.com: domain of sfitallstudentadmin@sfit.ac.in designates 209.85.220.41 as permi
Received-SPF: pass (google.com: domain of sfitallstudentadmin@sfit.ac.in designates 209.85.220.41 as pe
header.b=000LYCY5; arc=pass (i=1); spf=pass (google.com: domain
qnx7tIdJW1kSP qcCtuLc/NkAgHkEjkUc2lxnR5pKwoNQ1K/wBRv7K3mtIZLARJK7ws6vK6hsBd
v9sY71qpLcf2j5P2PbZ+TvE/mL/rM4KbAM8DnPSsfVYdQTxhpmoW+l3F3a29ncR SPF JEDukaMqAH
```

Observation:

The extracted authentication results contain SPF pass entries, indicating that the displayed SPF checks passed for the relevant sending domain.

Step 12: Check DKIM Status

```
(student@Kali)-[~/Shivam/email_forensics]
$ grep -i "dkim" evidence/suspicious_email.eml
:mime-version:dkim-signature;
dkim=pass header.i=@sfit-ac-in.20251104.gappssmtp.com header.s=20251104 header.b=DeGvhUmZ;
arc=pass (i=3 spf=pass spf-domain=sfit.ac.in dkim=pass dkdomain=sfit-ac-in.20251104.gappssmt
dkim=pass header.i=@sfit-ac-in.20251104.gappssmtp.com header.s=20251104 header.b=DeGvhUmZ;
arc=pass (i=3 spf=pass spf-domain=sfit.ac.in dkim=pass dkdomain=sfit-ac-in.20251104.gappssmt
:mime-version:dkim-signature;
dkim=pass header.i=@sfit-ac-in.20251104.gappssmtp.com header.s=20251104 header.b=000LYCY5;
DKIM-Signature: v=1; a=rsa-sha256; c=relaxed/relaxed;
X-Google-DKIM-Signature: v=1; a=rsa-sha256; c=relaxed/relaxed;
h=to:subject:message-id:date:from:mime-version:dkim-signature;
dkim=pass header.i=@sfit-ac-in.20251104.gappssmtp.com header.s=20251104 header.b=000LYCY5;
h=to:subject:message-id:date:from:mime-version:dkim-signature;
X-Original-Authentication-Results: mx.google.com; dkim=pass
```

Observation:

The extracted header information displays dkim=pass, indicating that the email's DKIM authentication result is reported as a pass.

Step 13: Check DMARC Status

```
(student@Kali)-[~/Shivam/email_forensics]
$ grep -i "dmarc" evidence/suspicious_email.eml
arc=pass (i=3 spf=pass spfdomain=sfit.ac.in dkim=pass dkdomain=sfit-ac-in.20251104.gappssmt;
dmarc=pass (p=NONE sp=NONE dis=NONE) header.from=sfit.ac.in;
arc=pass (i=3 spf=pass spfdomain=sfit.ac.in dkim=pass dkdomain=sfit-ac-in.20251104.gappssmt;
dmarc=pass (p=NONE sp=NONE dis=NONE) header.from=sfit.ac.in;
dmarc=pass (p=NONE sp=NONE dis=NONE) header.from=sfit.ac.in;
dmarc=pass (p=NONE sp=NONE dis=NONE) header.from=sfit.ac.in;
sender) smtp.mailfrom=sfitallstudentadmin@sfit.ac.in; dmarc=pass
```

Observation:

The extracted header contains DMARC authentication results reported as dmarc=pass, while the displayed policy fields include p=NONE, sp=NONE, and dis=NONE.

Step 14: Display Authentication Results

```
(student@Kali)-[~/Shivam/email_forensics]
$ grep -iE "spf|dkim|dmarc|Authentication-Results" evidence/suspicious_email.eml
:mime-version:dkim-signature;
ARC-Authentication-Results: i=4; mx.google.com;
dkim=pass header.i=@sfit-ac-in.20251104.gappssmt.com header.s=20251104 header.b=DeGvhUmZ;
arc=pass (i=3 spf=pass spfdomain=sfit.ac.in dkim=pass dkdomain=sfit-ac-in.20251104.gappssmt.com dmarc=pass (p=NONE sp=NONE dis=NONE) header.from=sfit.ac.in;
spf=pass (google.com: domain of sfitallstudent+bncbc4opa4k7ukrbqyg7dkqmgqez4lmumq@sfit.ac.in designates 209.85.220.41 as permitted);
MGQEZ4LMUMQ@sfit.ac.in;
dmarc=pass (p=NONE sp=NONE dis=NONE) header.from=sfit.ac.in;
Received-SPF: pass (google.com: domain of sfitallstudent+bncbc4opa4k7ukrbqyg7dkqmgqez4lmumq@sfit.ac.in designates 209.85.220.41 as permitted);
Authentication-Results: mx.google.com;
dkim=pass header.i=@sfit-ac-in.20251104.gappssmt.com header.s=20251104 header.b=DeGvhUmZ;
arc=pass (i=3 spf=pass spfdomain=sfit.ac.in dkim=pass dkdomain=sfit-ac-in.20251104.gappssmt.com dmarc=pass (p=NONE sp=NONE dis=NONE) header.from=sfit.ac.in;
spf=pass (google.com: domain of sfitallstudent+bncbc4opa4k7ukrbqyg7dkqmgqez4lmumq@sfit.ac.in designates 209.85.220.41 as permitted);
MGQEZ4LMUMQ@sfit.ac.in;
dmarc=pass (p=NONE sp=NONE dis=NONE) header.from=sfit.ac.in;
:mime-version:dkim-signature;
ARC-Authentication-Results: i=3; mx.google.com;
dkim=pass header.i=@sfit-ac-in.20251104.gappssmt.com header.s=20251104 header.b=000LYCY5;
spf=pass (google.com: domain of sfitallstudentadmin@sfit.ac.in designates 209.85.220.41 as permitted);
dmarc=pass (p=NONE sp=NONE dis=NONE) header.from=sfit.ac.in;
DKIM-Signature: v=1; a=rsa-sha256; c=relaxed/relaxed;
:mailing-list:precedence:x-original-authentication-results
X-Google-DKIM-Signature: v=1; a=rsa-sha256; c=relaxed/relaxed;
:x-original-authentication-results:x-original-sender:content-type:to
h=to:subject:message-id:date:from:mime-version:dkim-signature;
ARC-Authentication-Results: i=2; mx.google.com;
dkim=pass header.i=@sfit-ac-in.20251104.gappssmt.com header.s=20251104 header.b=000LYCY5;
spf=pass (google.com: domain of sfitallstudentadmin@sfit.ac.in designates 209.85.220.41 as permitted);
dmarc=pass (p=NONE sp=NONE dis=NONE) header.from=sfit.ac.in;
Received-SPF: pass (google.com: domain of sfitallstudentadmin@sfit.ac.in designates 209.85.220.41 as permitted);
```

Observation:

The combined search displays Authentication-Results and ARC-Authentication-Results information containing SPF, DKIM, and DMARC authentication results.

Step 15: Search for URLs

15.1 Extract URLs from the Raw Email

```
(student@Kali)~[~/Shivam/email_forensics]
$ grep -Eo 'https?://[^\s:]+ ' evidence/suspicious_email.eml
https://groups.google.com/a/sfit.ac.in/group/sfitallstudent/post>,
https://support.google.com/a/sfit.ac.in/bin/topic.py?topic=25838>,
https://groups.google.com/a/sfit.ac.in/group/sfitallstudent/>
https://groups.google.com/a/sfit.ac.in/group/sfitallstudent/subscribe>
```

15.2 Discover URLs Using String Analysis

```
(student@Kali)~[~/Shivam/email_forensics/analysis/attachments]
$ strings ../../evidence/suspicious_email.eml | grep http
List-Post: <https://groups.google.com/a/sfit.ac.in/group/sfitallstudent/post>, <mailto:sfitallstudent@sfit.ac.in>
List-Help: <https://support.google.com/a/sfit.ac.in/bin/topic.py?topic=25838>, <mailto:sfitallstudent+help@sfit.ac.in>
List-Archive: <https://groups.google.com/a/sfit.ac.in/group/sfitallstudent/>
<https://groups.google.com/a/sfit.ac.in/group/sfitallstudent/subscribe>
```

Observation:

- The URL extraction command identifies HTTP and HTTPS links within the email, including Google Groups URLs.
- The strings and grep http commands reveal additional readable URL references within the email's contents and headers.

Step 16: Search for Suspicious Keywords

```
(student@Kali)~[~/Shivam/email_forensics]
$ grep -iE "urgent|verify|password|account|payment|click|login" evidence/suspicious_email.eml
```

Observation:

The command searches the email for suspicious terms such as urgent, verify, password, account, payment, click, and login, but no matching output is visible in the screenshot.

Step 17: Examine MIME Information

17.1 Examine Content-Type Headers

```
(student@Kali)~[~/Shivam/email_forensics]
$ grep -i "Content-Type:" evidence/suspicious_email.eml
Content-Type: multipart/mixed; boundary="000000000000885fc8065cc4b98b"
Content-Type: multipart/alternative; boundary="000000000000885fc4065cc4b989"
Content-Type: text/plain; charset="UTF-8"
Content-Type: text/html; charset="UTF-8"
Content-Type: application/pdf;
```

17.2 Examine Content-Disposition Header

```
(student@Kali)-[~/Shivam/email_forensics]
$ grep -i "^Content-Disposition:" evidence/suspicious_email.eml
Content-Disposition: attachment;
```

Observation:

- The Content-Type headers identify MIME content types such as multipart/mixed, multipart/alternative, text/plain, text/html, and application/pdf.
- The Content-Disposition header displays attachment, indicating that the email contains an attachment.

Step 18: Extract Attachments

18.1 Install the mpack Utility

```
(student@Kali)-[~/Shivam/email_forensics]
$ sudo apt install mpack
[sudo] password for student:
Sorry, try again.
[sudo] password for student:
Installing:
 mpack

Suggested packages:
 default-mta | mail-transport-agent inews

Summary:
 Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 2744
 Download size: 37.6 kB
 Space needed: 112 kB / 7,530 MB available

Get:1 http://kali.download/kali kali-rolling/main amd64 mpack amd64 1.6-19 [37.6 kB]
Fetched 37.6 kB in 1s (43.1 kB/s)
Selecting previously unselected package mpack.
(Reading database... 558232 files and directories currently installed.)
Preparing to unpack .../mpack_1.6-19_amd64.deb...
Unpacking mpack (1.6-19)...
Setting up mpack (1.6-19)...
Processing triggers for kali-menu (2026.1.5)...
Processing triggers for man-db (2.13.1-1)...
```

18.2 Create the Attachments Directory

```
(student@Kali)-[~/Shivam/email_forensics]
$ mkdir -p analysis/attachments
```

18.3 Enter the Attachments Directory

```
(student@Kali)-[~/Shivam/email_forensics]
$ cd analysis/attachments

(student@Kali)-[~/Shivam/email_forensics/analysis/attachments]
$ \
```

18.4 Extract Attachments Using munpack

```
(student@Kali)-[~/Shivam/email_forensics/analysis/attachments]
$ munpack ../../evidence/suspicious_email.eml
tempdesc.txt: File exists
HolidayXNoticeXX02.10.2026X-XGandhiXJayantiX.pdf (application/pdf)
```

18.5 Verify the Extracted Attachment Files

```
(student@Kali)-[~/Shivam/email_forensics/analysis/attachments]
$ ls -lah
total 52K
drwxrwxr-x 2 student student 4.0K Oct  9 12:49 .
drwxrwxr-x 3 student student 4.0K Oct  9 12:48 ..
-rw----- 1 student student 41K Oct  9 12:49 HolidayXNoticeXX02.10.2026X-XGandhiXJayantiX.pdf
```

Observation:

- The terminal output shows the installation of the mpack utility required for attachment extraction.
- The mkdir -p analysis/attachments command creates the directory for storing extracted attachments.
- The cd analysis/attachments command changes the working directory to the attachments folder.
- The munpack command extracts the email attachment and identifies a PDF file; the output also indicates that tempdesc.txt already exists.
- The ls -lah command lists the extracted PDF file and displays its file details.

Step 19: Verify Attachment Type

```
(student@Kali)-[~/Shivam/email_forensics/analysis/attachments]
$ file HolidayXNoticeXX02.10.2026X-XGandhiXJayantiX.pdf
HolidayXNoticeXX02.10.2026X-XGandhiXJayantiX.pdf: PDF document, version 1.4, 1 page(s)
```

Observation:

The file command identifies the extracted attachment as a PDF document, version 1.4, containing one page.

Step 20: Analyze Metadata

20.1 Install ExifTool

```
(student@Kali)-[~/Shivam/email_forensics/analysis/attachments]
$ sudo apt install exiftool
Note, selecting 'libimage-exiftool-perl' instead of 'exiftool'
Upgrading:
  libimage-exiftool-perl

Summary:
  Upgrading: 1, Installing: 0, Removing: 0, Not Upgrading: 2743
  Download size: 6,106 kB
  Space needed: 869 kB / 7,530 MB available

Get:1 http://http.kali.org/kali kali-rolling/main amd64 libimage-exiftool-perl all 13.55+dfsg-1 [6,106 kB]
Fetched 6,106 kB in 2s (2,530 kB/s)
(Reading database... 558241 files and directories currently installed.)
Preparing to unpack .../libimage-exiftool-perl_13.55+dfsg-1_all.deb...
Unpacking libimage-exiftool-perl (13.55+dfsg-1) over (13.50+dfsg-1)...
Setting up libimage-exiftool-perl (13.55+dfsg-1)...
Processing triggers for doc-base (0.11.2)...
Processing 1 changed doc-base file...
Processing triggers for man-db (2.13.1-1)...
Processing triggers for kali-menu (2026.1.5)...
```

20.2 Examine Attachment Metadata

```
(student@Kali)-[~/Shivam/email_forensics/analysis/attachments]
$ exiftool HolidayXNoticeXX02.10.2026X-XGandhiXJayantiX.pdf
ExifTool Version Number      : 13.55
File Name                    : HolidayXNoticeXX02.10.2026X-XGandhiXJayantiX.pdf
Directory                    : .
File Size                    : 42 kB
File Modification Date/Time   : 2026:10:09 12:49:08+05:30
File Access Date/Time        : 2026:10:09 12:56:02+05:30
File Inode Change Date/Time   : 2026:10:09 12:49:08+05:30
File Permissions              : -rw-----
File Type                    : PDF
File Type Extension          : pdf
MIME Type                    : application/pdf
PDF Version                  : 1.4
Linearized                   : No
Create Date                  : 2026:10:01 15:40:15+05:30
Creator                     : NAPS2
Author                      : NAPS2
Subject                     : Scanned Image
Title                      : Scanned Image
Modify Date                  : 2026:10:01 15:40:15+05:30
Producer                    : PDFsharp 1.50.4000-netstandard (https://github.com/ststeiger/PdfSharpCore)
Page Count                   : 1
```

Observation:

- The terminal output shows the installation or update of the ExifTool package and its associated dependency.
- ExifTool identifies the attachment as a PDF file and displays metadata such as file size, PDF version, page count, creator, author, subject, and modification date.

Step 21: Extract Readable Strings

```
(student@Kali)-[~/Shivam/email_forensics/analysis/attachments]  
$ strings HolidayXNoticeXX02.10.2026X-XGandhiXJayantiX.pdf | less
```

```
%PDF-1.4  
1 0 obj  
/CreationDate(D:20261001154015+05'30')  
/Creator(NAPS2)  
/Author(NAPS2)  
/Keywords()  
/Subject(Scanned Image)  
/Title(Scanned Image)  
/ModDate(D:20261001154015+05'30')  
/Producer(PDFsharp 1.50.4000-netstandard \(\https://github.com/ststeiger/PdfSharpCore\))  
endobj  
2 0 obj  
/Type/Catalog  
/Pages 3 0 R  
endobj  
3 0 obj  
/Type/Pages  
/Count 1  
/Kids[4 0 R]  
endobj  
4 0 obj  
/Type/Page  
/MediaBox[0 0 599.04 852.48]  
/Parent 3 0 R  
/Contents 5 0 R  
/Resources  
/ProcSet [/PDF/Text/ImageB/ImageC/ImageI]  
/ExtGState  
/GS0 6 0 R  
/XObject  
/I0 7 0 R  
/Group  
/CS/DeviceRGB  
/S/Transparency  
endobj  
5 0 obj  
/Length 57
```

Observation:

- The strings command is executed on the extracted PDF using less to view its readable contents.
- The output displays readable PDF structure and metadata, including creator and producer information and an embedded URL.

Step 22: Build an Email Timeline

```
(student@Kali)-[~/Shivam/email_forensics/analysis/attachments]
$ grep -iE "^(Date|Received):" ../../evidence/suspicious_email.eml
Received: by 2002:a05:6504:e081:20b0:314:537d:9a10 with SMTP id m1-n2csp2146006ltd;
Received: from mail-sor-f69.google.com (mail-sor-f69.google.com. [209.85.220.69])
Received: by 2002:a17:902:e80d:b0:2cc:89b8:4246 with SMTP id
Received: from mail-sor-f41.google.com (mail-sor-f41.google.com. [209.85.220.41])
Date: Thu, 1 Oct 2026 15:47:24 +0530
```

Observation:

The command extracts the Date and Received headers together, displaying the message timestamp and mail server routing information for reconstructing the email timeline.